Avocado Inspection Manual

“Quality Control Service at Destination”

AVO-INS-MAN-DES-02

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Index

1. Glossary 3
2. Description & Sampling 4
3. Inspection Procedure & Inspection Instruments 6
4. Standard Avocado 10
5. Quality Defects 11
6. Condition Defects 17

1. Glossary

- ✓ **Box Label Information:** Information present in every box which include specific information about the fruit such as variety, size, packing date, grower, category.
- ✓ **Commodity:** This term is used as a synonym for species. Ex: Apples, Cherries, Oranges, etc.
- ✓ **Condition defects:** All damage that has evolution over time, due to temperature, duration of the transportation, etc. Ex: Open wounds, putrefaction, damages, etc.
- ✓ **Container:** is a cargo recipient used in sea, terrestrial or multimodal transportation. For our interests, it is used to transport fruit and vegetables from the loading center to its destination.
- ✓ **Container number:** Each container has an individual number that identifies it. The numbers are composed of 4 letters, 6 numbers and an identification digit (ex: SUDU 123456-7).
- ✓ **Lamp (Ventilation):** Is the ventilation regulator that is found in the posterior part of the container which allows the gas exchange of the environmental atmosphere in the inside of the container.
- ✓ **Package/box:** What protects the fruit during the trip, it can be of paper, plastic or wood.
- ✓ **Packing list:** List corresponding to a specific container that indicated the pallets per producer, size and number of boxes.
- ✓ **Pallet:** Set of boxes packed and consolidated in a pallet base, labelled with a card and sheets.
- ✓ **Pallet Number:** Internal and self-copying register that identifies and individualizes the information of a pallet within a list.
- ✓ **Quality defects:** All of the damage that does not have an evolution over time, where the grade of the damage does not vary. For example: size, presence of foreign elements, healed wounds, etc.
- ✓ **Sample:** Selection of boxes within the container which should be inspected, whenever possible, that will cover all of the distinct parameters such as varieties, packing date, and producers, with the objective that the sample will be the most representative possible.
- ✓ **Thermograph:** Sensor that registers the temperature of the container from the origin to destination.
- ✓ **Ethylene**: Ethylene is a phytohormone in gaseous form synthesized naturally by fruits and vegetables. It is responsible of regularizing maturity, senescence and related physiological processes.
- ✓ **Lenticels:** The lenticels are pores in the outher skin that allows the direct interchange of gases between the internal and the atmosphere.

2. Description & Sampling

PERSEA AMERICANA is a tropical tree specie, which fruit “Avocado”, is a edible drupe. The avocado is a climacteric fruit, which means it will get mature after harvest and as a result gets soft and usually undergoes a change of color to black. Some commercial varieties are: Hass, Fuerte, Ettinger, Zutano, etc.





Sampling

Before performing the inspection be aware of the different parameters present on the container / lot such as:

- Variety
- Packing Date
- Grower
- Size
- Category (CAT1)
- Packaging

The parameters mentioned above can be present within the same container / lot and the different combinations that are present must be taken representatively. This information can be found in the Packing List, this document will show the detail of the content of the container / lot and depending on the information available on the packing list, the number of parameters to be aware of at the sample selection. The main objective of a good sampling is to ensure that the inspection is carried out with a good representativeness on container / lot.

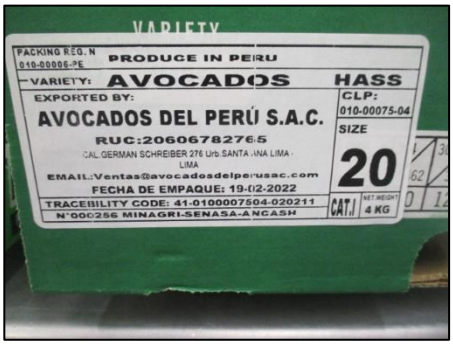
The amount of boxes to select for the sample in a regular inspection is between 8 – 10 boxes from different pallets considering different varieties, growers, packing dates, present label.

There are different packaging formats, the most 2 common format packagings are the following:

- 4kg Carton box packaging:

The amount of avocados inside the box should be the same as the size indicated in the label, the net weigh of the fruit should be above 4kg and the avocado fruits have a weight corresponding to the size, depending on the exporter indicated at the box information label. If the size is lower than 24 (24 units in a box) at least 2 boxes should be selected from the pallet to complete one sample.





- 10kg Plastic crate packaging:

Avocados packed in a 10kg plastic bulk packaging and according to the size, each avocado fruit should have the correspondent weight depending on the exporter indicated at the box information label. The net weight should be above 10kg. The amount of avocados inside the box will vary depending on the size of the fruit, the count of avocados in every sample must be counted to note the information during the inspection.





3. Inspection Procedure & Inspection Instruments

For a common avocado inspection 8 - 10 samples must be taken, taking into account the different sizes, varieties, grower, etc. Always take pulp temperature at the site where the fruit is stored or available for the inspection. The upper box of the pallet is recommendable not to be taken for inspection as that one, during the trip, is more susceptible to deteriorate due to the temperature, ventilation, etc, so for the sample selection take the boxes from the second down. Take the net weight and while making the inspection, identifying and separating the defects found, count the avocados inside the box as this is the 100% of the sample. Take the correspondent measurements during the inspection such as pulp temperature, cut pulp and firmness. For every sample take photos of the pallet number label, box information label and all the inspection process.





The sample of avocados to have a good general appearance should have homogeneous green color and size, absence of defects and clean. The picture showing the appearance of the fruit has to be always present on the report pictures.





Inspection Instruments

Weighing scale

Instrument used to measure te weight of the sample or fruit units. The net weight of the sample should be superior to the weight mentioned on the label also the weight of the avocado fruit should be in accordance with the size, below an example of the weight distribution per size, this can vary among the exporters.

Weight scale in grammes	Size code
781 to 1 220	4
576 to 780	6
461 to 575	8
366 to 460	10
306 to 365	12
266 to 305	14
236 to 265	16
211 to 235	18
191 to 210	20
171 to 190	22
156 to 170	24
146 to 155	26
136 to 145	28
125 to 135	30
80 to 125 (Hass variety only)	S (The difference between the smallest and largest fruit within a package should not be more than 25 g.)





Temperature

Instrument used to measure the temperature of the fruit or environmental temperature.

The pulp temperature measurement has to be taken at the arrival of the container to the destination warehouse or at the site where the fruit is stored, for example a cold room. Always have in consideration take the environmental temperature.

During the inspection also is important to measure the pulp temperature to every sample.

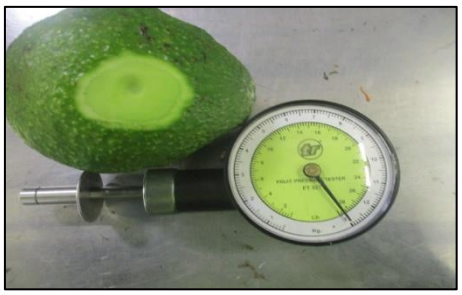




Fruit Firmness

The fruit is expected to be firm as a stone at the moment of arrival at the warehouse, note that if the firmness don't score the maximum it means the avocado is already getting softer.

The firmness has to be measured at the equatorial part of the avocado fruit, cutting the skin with the peeler and applying pressure directly to the pulp.





4. Standard Avocado

Parameters/Tolerances	Very Good 1	Good 2	Fair 3	Poor 4	Bad 5
Quality Defects	%	%	%	%	%
General Appearance	1	2	3	4	5
Color Turning	0 - 5	>5 - 15	> 15 - 30	> 30 - 45	> 45
Color Black	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Foreign Material / Dirty	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Misshapen	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Scars / Russet Severe	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Scars / Russet Slight / Moderate	0 - 5	>5 - 15	> 15 - 30	> 30 - 45	> 45
Skin Abrassions	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Trips Damage	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Scales	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Sunburn	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Rough Skin	0 - 5	>5 - 15	> 15 - 30	> 30 - 45	> 45
Missize	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Torn Stem-end	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Long Stalk	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Quimical Injury	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Insect Presence / Damage	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
TOTAL QUALITY DEFECTS	**0 - 5**	**>5 - 15**	**> 15 - 30**	**> 30 - 45**	**> 45**
Condition Defects	%	%	%	%	%
Decay	0	> 0 - 4	> 4 - 6	> 6 - 10	> 10
Stem End Rot	0	> 0 - 5	> 5 - 10	> 10 - 25	> 25
Mold	0	> 0 - 4	> 4 - 6	> 6 - 10	> 10
Pedicelar Mold	0	> 0 - 5	> 5 - 10	> 10 - 25	> 25
Open Wounds	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Internal Breakdown	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Rodent Damage	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Bruises	0	> 0 - 5	> 5 - 10	> 10 - 25	> 25
Chilling / Freezing Damage	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Lenticelosis	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Anthracnose Severe	0	> 0 - 5	> 5 - 10	> 10 - 25	> 25
Anthracnose Slight / Moderate	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
Soft Fruit / Overripe	0	> 0 - 5	> 5 - 10	> 10 - 25	> 25
Black Spot	0 - 5	> 5 - 10	> 10 - 25	> 25 - 45	> 45
TOTAL CONDITION DEFECTS	**0 - 5**	**> 5 - 10**	**> 10 - 25**	**> 25 - 45**	**> 45**

5. Quality Defects

All of damage that does not have an evolution over time, where the grade of the damage does not vary. For example: size, presence of foreign elements, healed wounds, etc. All the parameters that don't have evolution over time, usually comes from the pre-harvest period, harvest and packinghouse handling.

Color turning

Color turning is a intermediate stage between the green color fruit and the black color fruit. (Figure 3 – 4)



Color Black

Color turning from dark or black color indicating that the fruit is ripen. (Figure (5).



This defects refers only to the color of the avocado as in some cases are ripe when these are full green in color or black color and they should be considered as soft fruit.

Foreign Material / Dirty (Unclean)

Fruit with plant remains or non-plant residues material attached to the fruit because of insufficient washing or sorting after harvesting.





Misshapen

Malformation of the fruit, where a slight malformation is allowed.





Scars / Russet Severe

Serious superficial damage of the skin mostly due to rubbing or hits in the stages of harvest or port-harvest. Scarring, rubbing and scratching falls under the same defect. The Scars / Russet can be considered as a defect when the surface of the damaged área is superior to 3 – 6cm2.



Scars / Russet Slight / Moderate

Slight to Moderate superficial damage of the skin mostly due to rubbing or hits in the stages of harvest or port-harvest. Scarring, rubbing and scratching falls under the same defect. The Scars / Russet can be considered as a defect when the surface of the damaged área is superior to 3 – 6cm2.



Skin Abrasions

Superficial damage of the skin with a corky surface, mostly due to rubbing.





Trips Damage

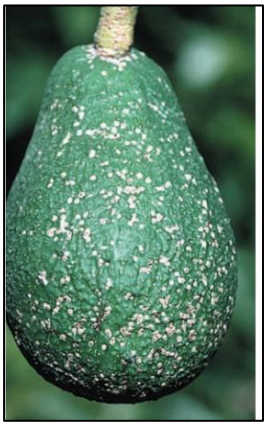
Caractherized for a pale Brown colouring on the skin caused by Thrips and small black spots excreteds by the insect. Although the presence of scales on the skin of fruit does not cause internal damage, it may lead to rejection of Fruit



Scales

Scales are small insects occasionally found attached to the fruit. Although the presence of scales on the skin of fruit does not cause internal damage, it may lead to rejection of Fruit.





Sun Burn

Brownish-yellowish spots on fruit surface always occurring on the outside of the canopy. a very slight skin defect caused by sunburn is tolerated, necrotic stains which are not allowed.





Rough Skin

Considered as a defect if the fruit has a very rough skin, mostly for Chinese market this defect has more importance.



Missize

Defect considered when the avocado fruit has a weight that don't correspond to the size. Usually the underweight (undersize) is a defect that can be a cause of underweight of the sample.



Torn Stem-end

The stem end must be attached to the fruit but when it is not present is a sign of careless handling of the fruit during harvest and commercialization. Also the open wound left can be a cause of pedicelar mold.





Long Stalk

The stalk may also not be longer than 10mm, above that it can be considered as a defect.



Chemical Injury

Residues of chemical products applied, usually at the farm. In severe cases they can damage the surface of the skin but generally is washable not affecting the surface.



Insect / Pest Damage

Defect caused by attack of insects on early stages of growing and that can result on malformations. If there is presence of insects most probably they will be already dead cause the fumigation.



6. Condition Defects

All of damage that has evolution over time, due to temperature, duration and conditions of transportation, etc. and finally reduce shelf life of the fruit. For example: Open wounds, putrefaction, Soft Fruit, etc.

Decay

During the decay process, the fruit have a brown color, disintegration with an aqueous aspect and a fetid odor resulting from fermentation.



Stem End Rot

Stem end rot is an important post-harvest disease of avocado. This disease cannot be detected at the green stage and symptoms of the defect can be easily determined when the fruits ripen.





Mold

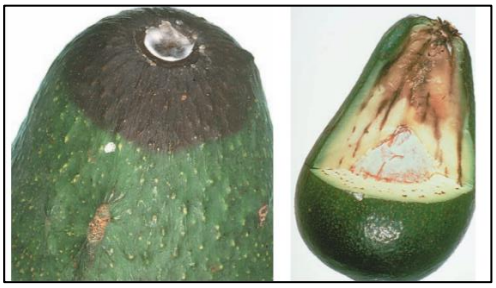
Mold presence on the fruit, in severe cases the fruit looks rotten.Mature fruit is more susceptible.





Pedicelar Mold

Mycelia present in the pedicle area, if the mold have the right conditions it can produce stem end rot.





Open Wounds

Unhealed wounds and injuries that leaves the flesh exposed to the environment making it more susceptible to develop mold or decay.




Internal Breakdown

The pulp turn to color gray or black due to several reasons such as temperature, nutritional or water deficiency, etc. This color change can also makes the flavour vary and the fruit look unappetizing.




Rodent Damage

Damage made by rodent, characterized by the long smooth tooth marks, usually related to a bad handling during the harvest and leaves the pulp exposed to the enviroment making it more propense to develop decay or mold.




Bruises

Rough handling or package pressure can cause bruising damage. Only a very slight bruise is allowed. Ripe avocados are more susceptible to get bruised spots.



Chilling / Freezing Damage

Due to cold temperature the chilling damage can be detected when the peel of the fruit shows dark brown to black discoloration. Internally and the fruit flesh may show a greyish discoloration. Chilling injury results in internal breakdown of the flesh. Is always important to check the temp logger for temperature measurement during transport.



Lenticelosis

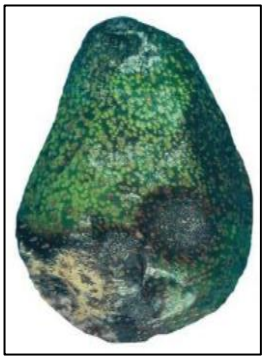
The lenticels damage can be seen as black spots that correspond to collapsed lenticels due to rough handling on the harvest, warehouse process line, brush cleaning, etc.

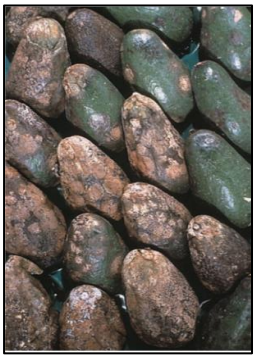




Anthracnose Severe

Anthracnose in an advanced stage and spread in a big area over the fruit and probably also spread to other neighbor fruits. This fungus also affects the pulp of the fruit.





Anthracnose Slight/Mod

Anthracnose is a fungal disease that appears located on the surface of the avocado and that in advanced stages can affect the pulp of the fruit. It can be seen in the beginning at small circular spots.

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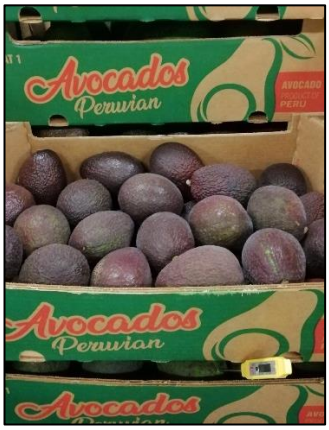




Soft Fruit/Overripe

As a climacteric fruit the avocado reaches the maturity after harvest showing, generally, change of color from green to black and also loosing the firmness till is ready to eat. The temperature, storage conditions, transit period can be related to the incidence of the soft fruit in a lot.





Black Spot

Damage caused by excess of CO2 % during the trip. It can be detected by incidence of brown color that can be located irregularly on the surface or cover completely the fruit and will not affect the pulp.

